Artificial Intelligence for Digital Characters

Autonomous Agents I

Chen Yang, ETH Zürich

Autonomous Agents

Dialogue Trees

Knowledge Graphs

Definition

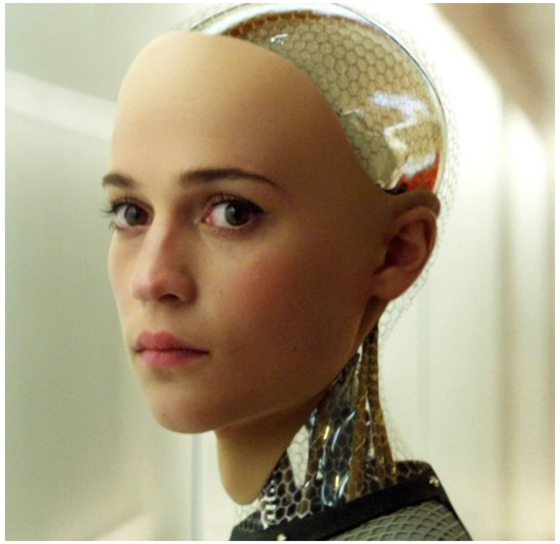
Characteristics

Functionality Modules

Definition

Agent

Anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.



Characteristics

Characteristics of Autonomous Agents

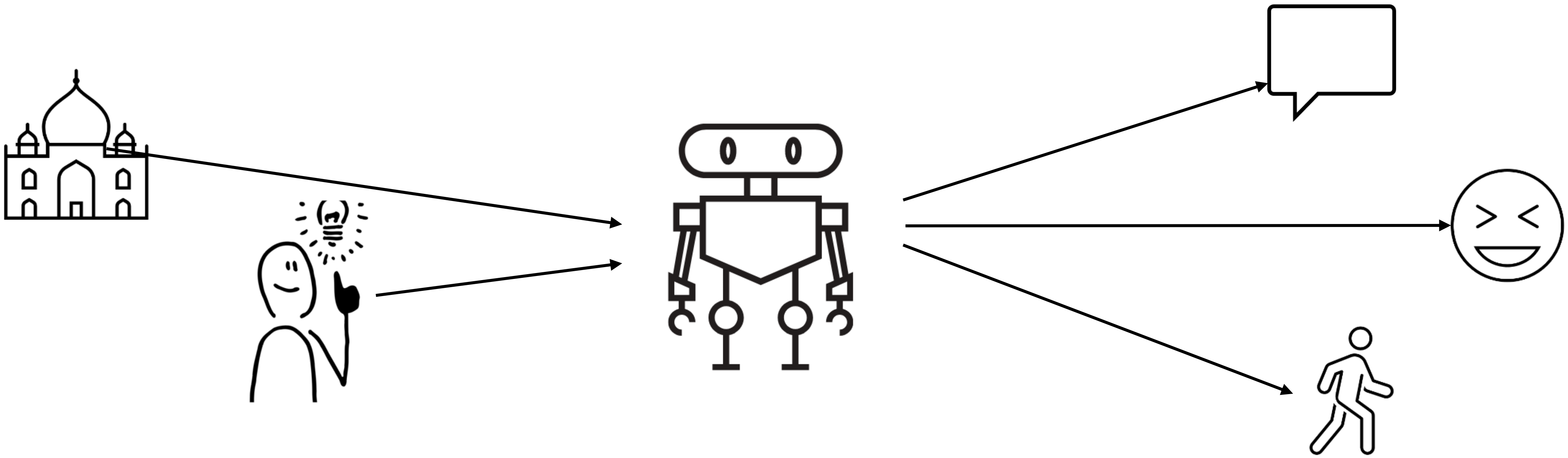
- **Reactivity**: perceive their environment and respond in a timely fashion to changes that occur in it.

- **Autonomy**: operate without direct intervention of humans and have control over their actions and internal states

- **Proactiveness**: plan for the future to take advantage of opportunities in the environment, to avoid future problems, deadlocks or to propagate changes or distribute information.

Digital Characters - Socially Intelligent Agents :

Interactive digital characters that exhibit human-like qualities and can communicate with humans and each other using natural human modalities like facial expressions, speech and gestures. They are capable of real-time perception, cognition, emotions, and action that allow them to participate in dynamic social environments with human-style intelligence.



Functionality Overview

Input

Audio/Video

Environment

Scene understanding
Object detection
Affordance recognition
...

Rapport Estimation

Accoustic Analysis
Facial Analysis
Guesture Analysis

Language

Language Recognition (EN)
Language Understanding

Processing

Planning

Learning

Explicit Knowledge representation
RL – Adaptive Agent

Memory

Environment
User model
Interaction history
Story consistency

Task Reasoner

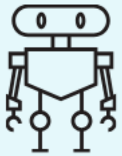
Agent Goal

Social Reasoner

Emotion
Personality
Empathy & Prosociality &Rapport
Culture
Theory of Mind

Output

Multi-modal Output



Assets augmentation

Remove objects from the scene
Add assets to the scene

Non-verbal behavior

Facial expression
Body language
Navigation & Locomotion
Posing in the environment

Expressive Speech

Language generation
Lip sync
Speech synthesis

Functionality Overview

Processing / Decision-Making:

- Core functionality of an autonomous agent
- Decision-making based on the **input** it receives from the environment, its **internal states** and its **goals**
- Common techniques:
    - **Finite State Machines**: a set of states and the conditions under which the agent transitions from one state to another.
    - **Behavior/Dialogue Trees**: an advancement over FSMs, allow for more complex decision-making by structuring decisions as a hierarchy of nodes that represent tasks or choices.
    - **Machine Learning/ Deep Learning** (such as Large Language Models)
    - ...

Functionality Overview

Processing / Decision-Making with LLMs:

Connect knowledge graphs to improve factual accuracy

Pros

- Continuous interaction between agents and their environment
- Task decomposition and planning
- Tool use and real-world Interaction

Cons

- Hallucination problem
- Adaption and learning challenges
- Planning and execution Limitations

Autonomous Agents

Dialogue Trees

Knowledge Graphs

Applications
Theoretical Overview
Software Ecosystem
Pros and Cons

Applications

Interactive Chat Experience

Games

Theoretical Overview

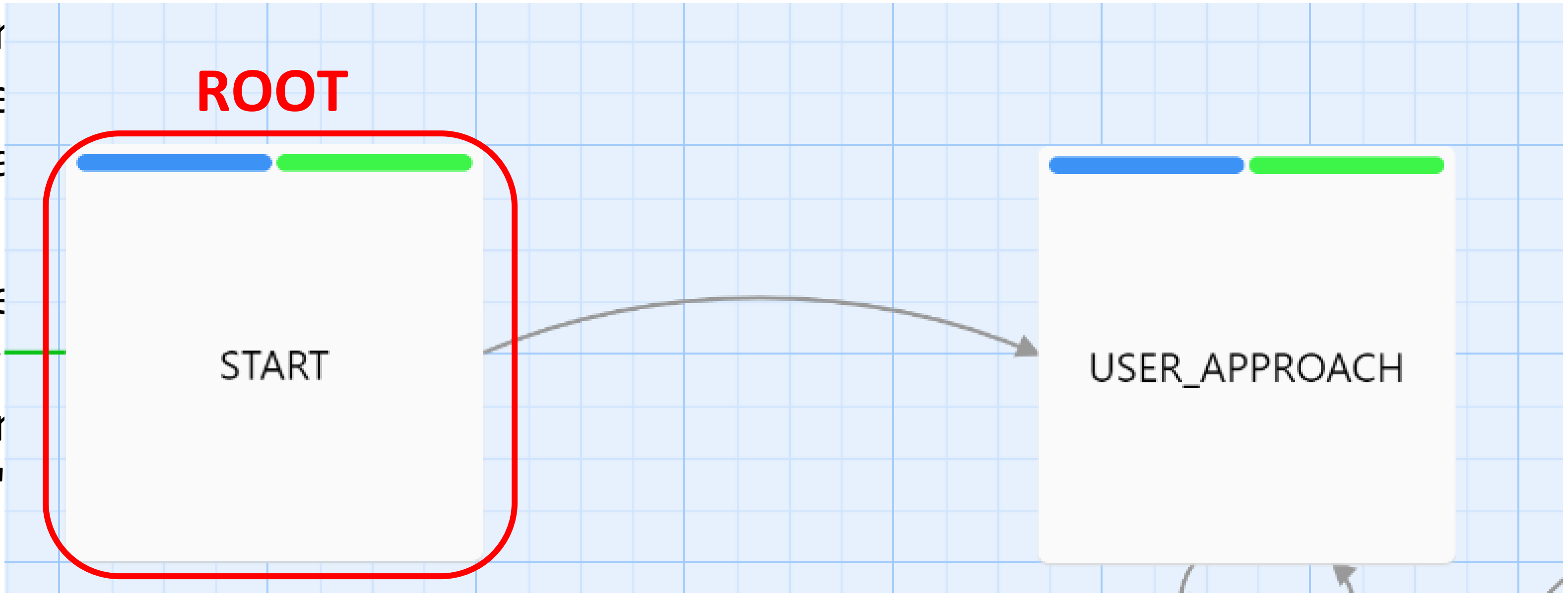
Main Concepts

Root: The starting point of the dialogue tree. This is the initial piece of dialogue or the first question asked.

Nodes: a piece of dialogue, a question, or a choice presented to the user

Branches: the connections between r
represent the user's possible response
the conversation can take following ea
point.

Leaves: the endpoints of the dialogue
conversation can conclude. These mig
end of a conversation, the achievemer
particular outcome based on the user'

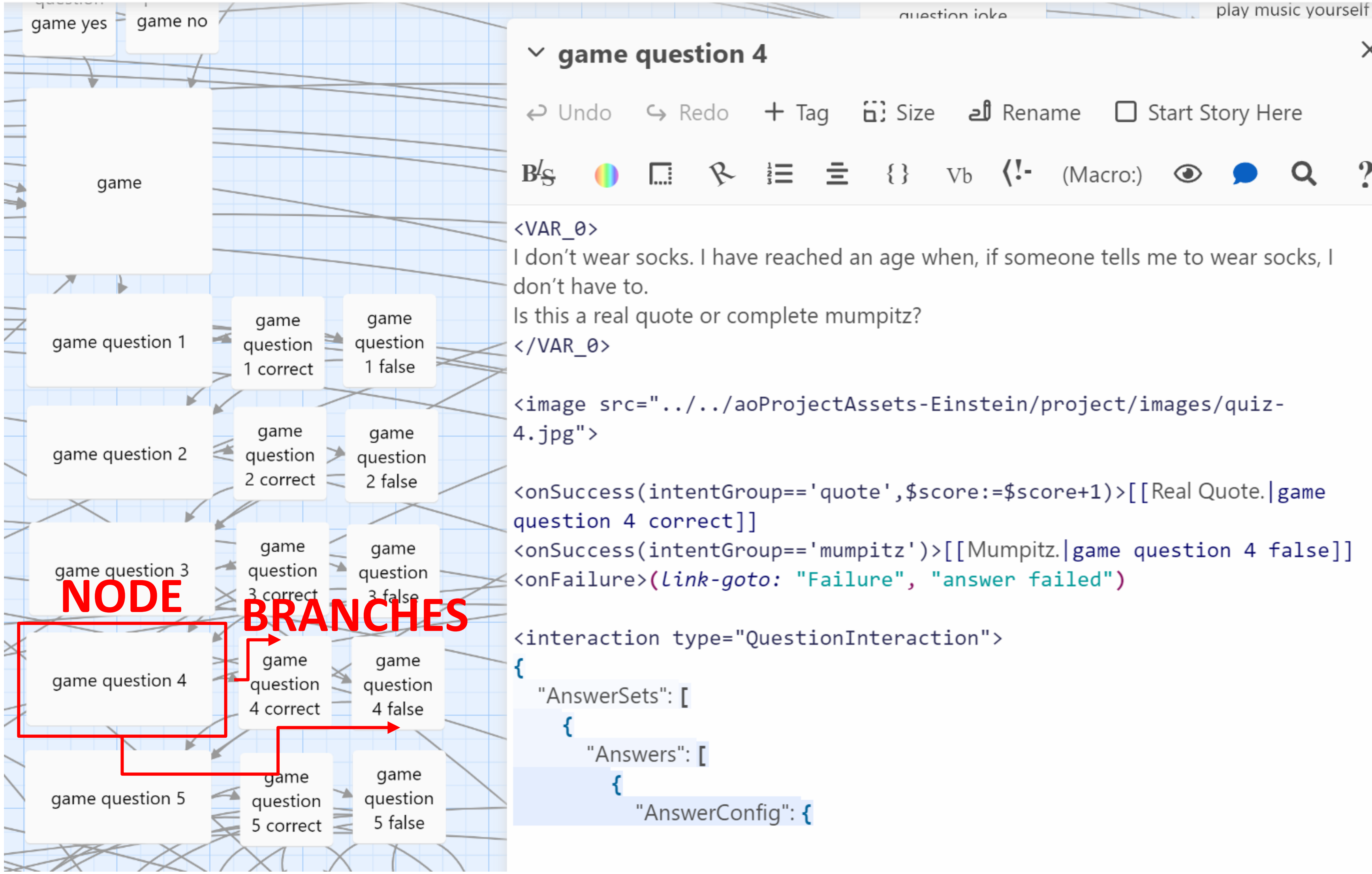


Theoretical Overview

Main Concepts

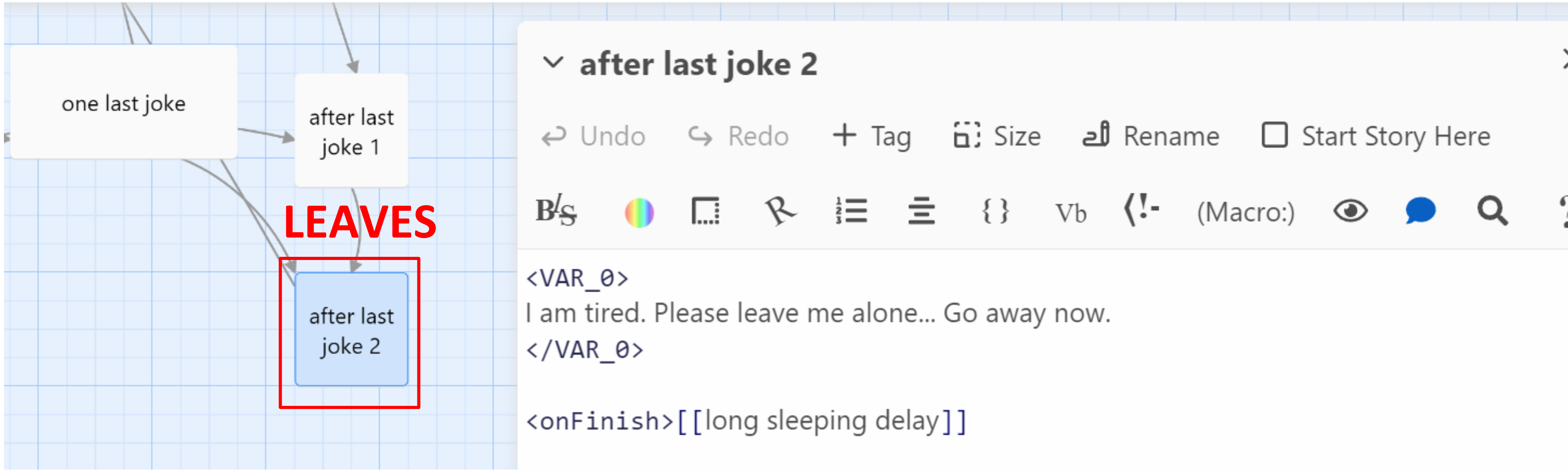
Nodes: a piece of dialogue, a question, or a choice presented to the user

Branches: the connections between nodes that represent the user's possible responses or the direction the conversation can take following each decision point.



Main Concepts

Leaves: The endpoints of the dialogue tree where the conversation can conclude. These might represent the end of a conversation, the achievement of a goal, or a particular outcome based on the user's choices.



Software and Platforms

Crafting Dialogue Trees

Twine: Open-source tool for telling interactive, nonlinear stories. No coding experience required. Directly publish to HTML.

Yarn Spinner: For interactive story telling, suitable for writers, programmers. Easy to connect with Unity.

Chat Mapper: For writing and testing nonlinear dialogue and events across various fields such as entertainment, e-learning, and emergency response

Conversational Agents with Dialogue Trees

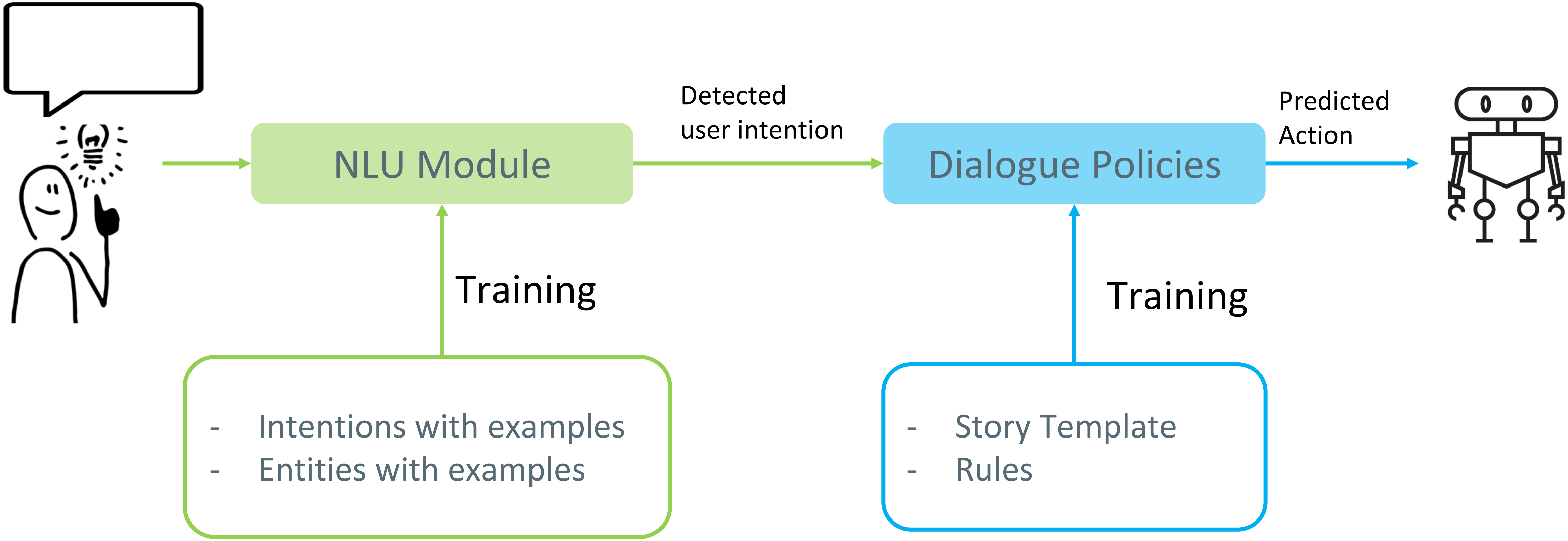
Rasa: Open-source machine learning framework for building AI-driven Chatbots and voice assistants. Offers granular control over the project.

Dialogue Flow: Backed by Google, offers user-friendly, integrated environment for developing chatbots and voice apps.

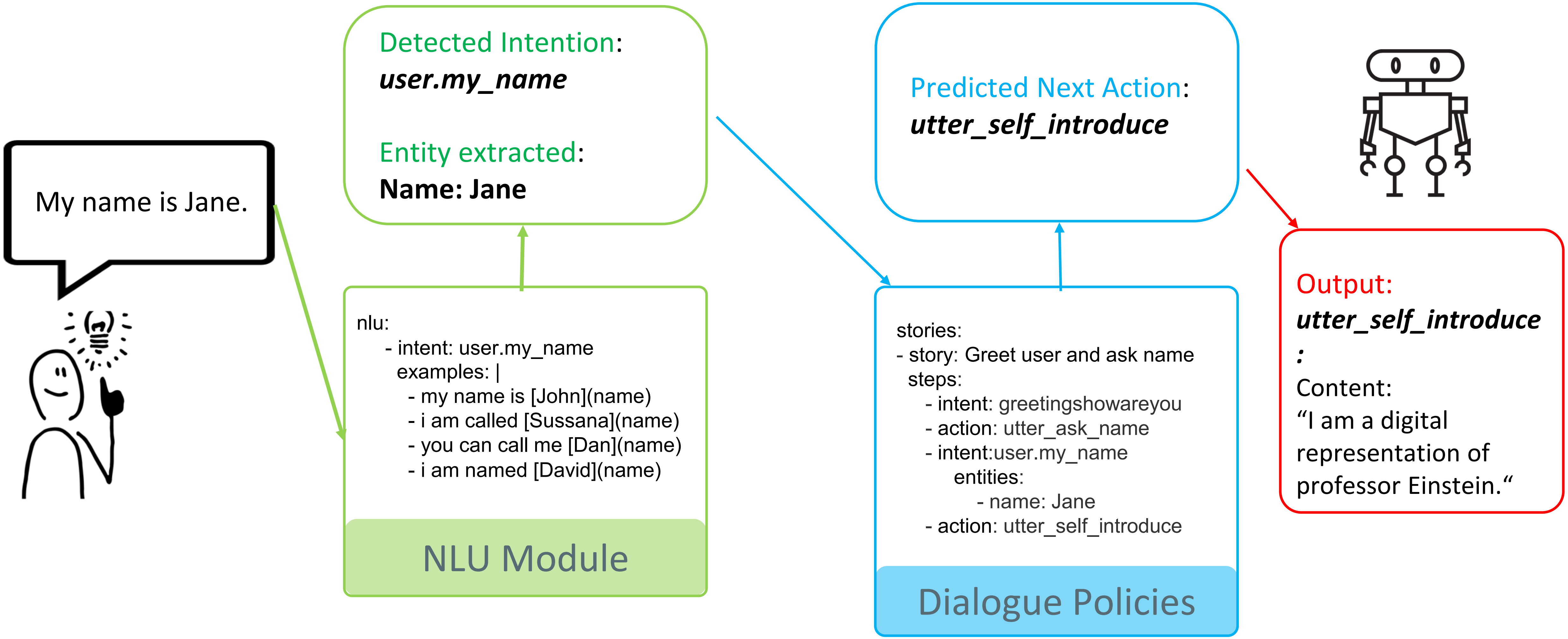
IBM Watson Assistant: Robust, enterprise-grade solution with advanced AI capabilities, mostly used for complex customer support applications and integrations with existing business systems.

Software and Platforms

RASA



RASA


My name is Jane.
Detected Intention:
user.my_name
Entity extracted:
Name: Jane
nlu:
- intent: user.my_name
examples: |
- my name is [John](name)
- i am called [Sussana](name)
- you can call me [Dan](name)
- i am named [David](name)
NLU Module
Predicted Next Action:
utter_self_introduce
stories:
- story: Greet user and ask name
steps:
- intent: greetingshowareyou
- action: utter_ask_name
- intent:user.my_name
entities:
- name: Jane
- action: utter_self_introduce
Dialogue Policies
Output:
utter_self_introduce
:
Content:
“I am a digital representation of professor Einstein.“

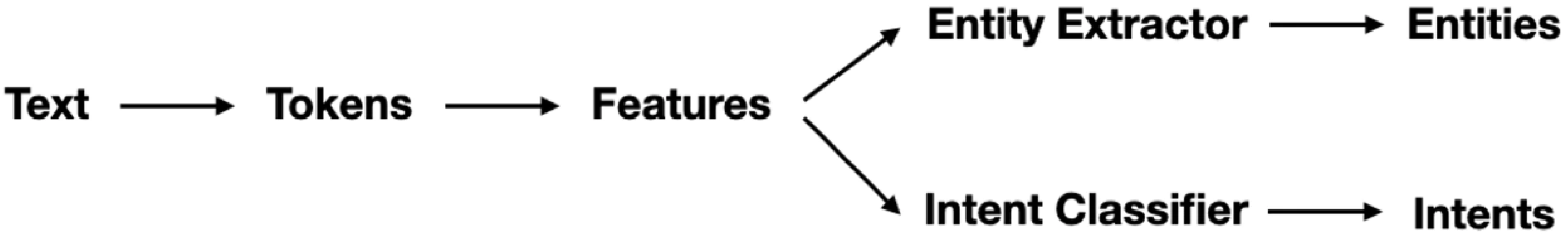
Software and Platforms

RASA – NLU Module

- Performs intent classification and entity extraction

- Rasa provide different components that can be directly configured in the NLU pipeline

```yaml
pipeline:
- name: WhitespaceTokenizer
- name: RegexFeaturizer
- name: LexicalSyntacticFeaturizer
- name: CountVectorsFeaturizer
- name: CountVectorsFeaturizer
  analyzer: char_wb
  min_ngram: 1
  max_ngram: 4
- name: DIETClassifier
  epochs: 100
```



Software and Platforms

RASA – NLU Module

Training

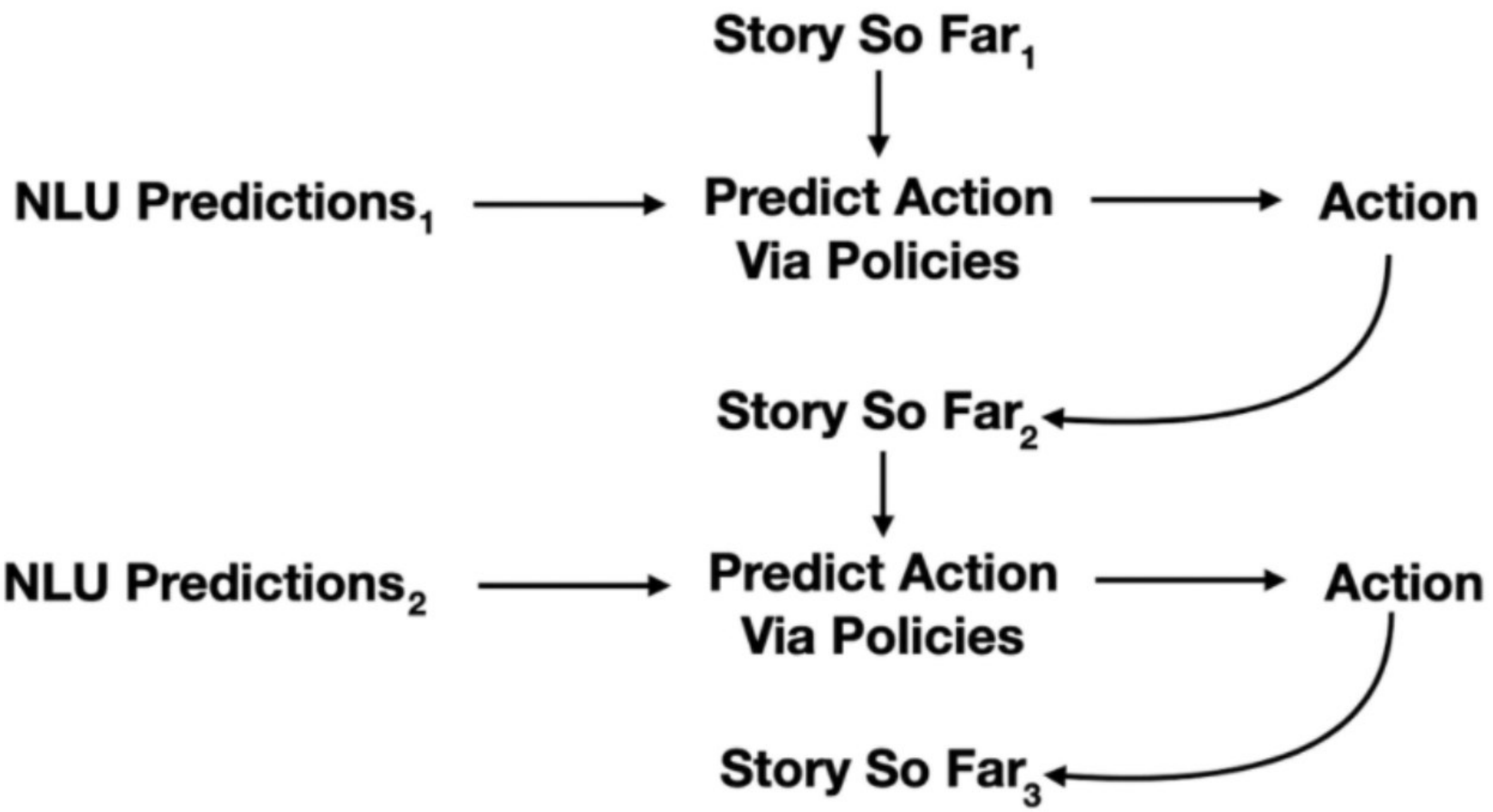
Training Data: structured information about user messages, YAML files.

Extra:

- Regular Expressions
- Lookup Tables


```yaml
- intent: user.my_name
  examples: |
    - my name is [John](name)
    - i am called [Sussana](name)
    - you can call me [Dan](name)
    - i am named [David](name)
    - please call me [Sarah](name)
    - people call me [Ben](name)
- intent: affirm
  examples: |
    - yes I do
    - yes please
    - Yes I'd love to
    - yes
    - of course
    - yes I think you passed
    - yes I have
    - yes I think so
    - yes I am
    - yes it is
    - yes that sounds good
    - yea
    - yes I am
    - yes I do
    - indeed
    - correct
```

RASA – Dialogue Policies



RASA – Dialogue Policies

Policy Configuration:

Machine Learning Policy:
- TED Policy
- Memorization Policy
- …

Rule-based Policies:
- Rule Policy

```yaml
policies:
  - name: MemoizationPolicy
    max_history: 1
    epochs: 500
    priority: 5
  - name: TEDPolicy
    max_history: 2
    epochs: 300
    constrain_similarities: True
    use_gpu: True
    priority: 1
  - name: RulePolicy
    core_fallback_threshold: 0.2
    epochs: 500
    core_fallback_action_name: "action_einstein_default_fallback"
    enable_fallback_prediction: True
    priority: 6
  - name: AugmentedMemoizationPolicy
    max_history: 2
    epochs: 300
    priority: 4
```

RASA – Dialogue Policies

Training:

Stories:
- TED Policy
- Memorization Policy
- …

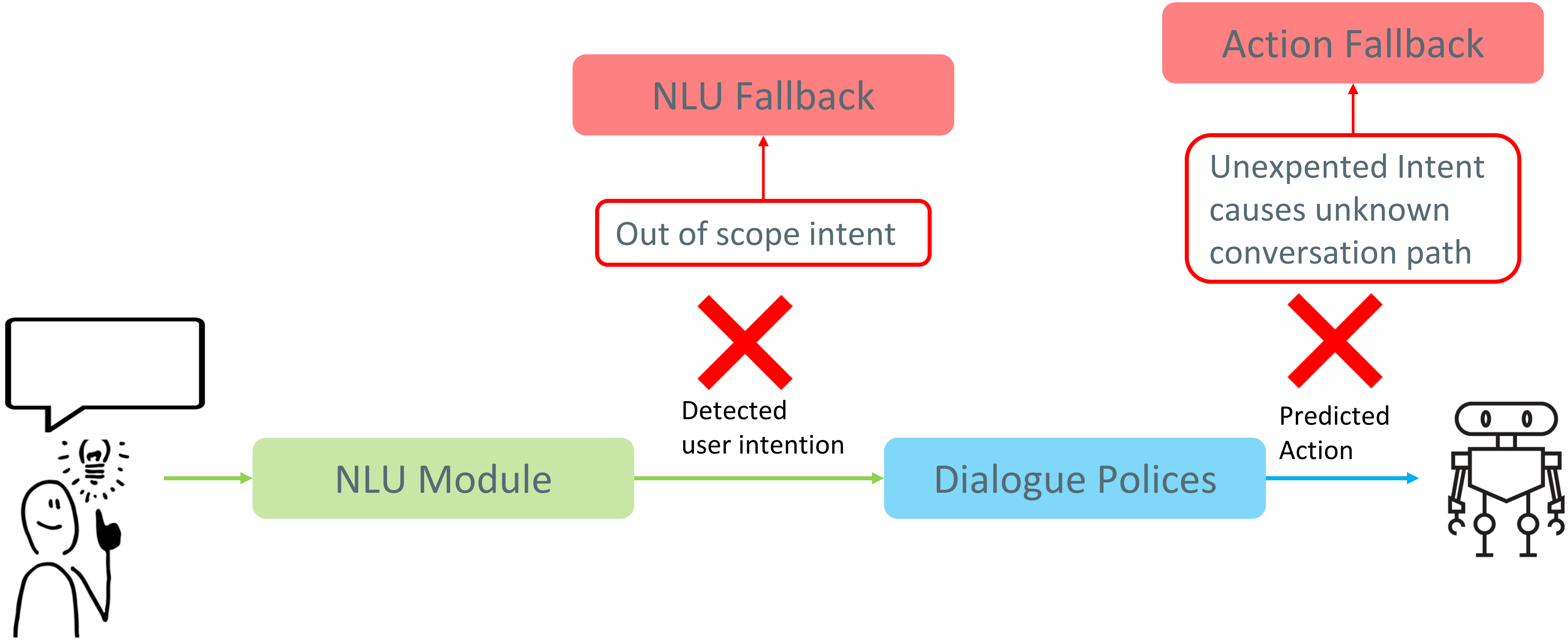
Rules:
- Rule Policy


```yaml
- story: interactive_story_2
  steps:
  - action: action_random_intro
  - intent: greetingshowareyou
  - action: action_utter_ask_name
  - intent: user.my_name
  - action: utter_self_introduce

- story: interactive_story_3
  steps:
  - action: action_random_intro
  - intent: greetingshowareyou
  - action: action_after_question_intention
  - intent: real human_yes
  - action: action_after_question_intention
  - intent: olympiaAcademia
```

Software and Platforms

RASA - Fallbacks



Pros and Cons

Advantages:

- **Predictability**: predefined responses
- **Simplicity:** easy to design and implement
- **Fast and Efficient:** inference time
- **Easier to Ensure Compliance**: follow story lines and avoid prohibited topics

Disadvantages:

- **Limited Flexibility**: hard to respond to unexpected queries or engage in open-ended conversations.
- **Poor Scalability:** to cover new topics or scenarios requires manual new design

Decision Making with LLMs

Connect knowledge graphs to improve factual accuracy

Pros

- Continuous interaction between agents and their environment
- Task decomposition and planning
- Tool use and real-world Interaction

Cons

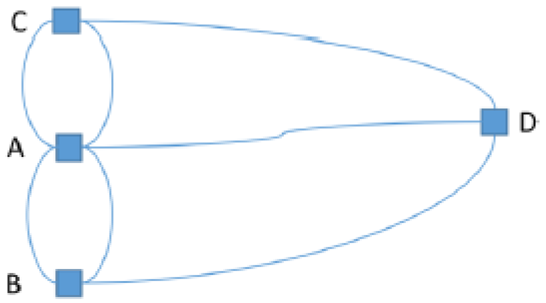
- Hallucination problem
- Adaption and learning challenges
- Planning and execution Limitations

Autonomous Agents
Dialogue Trees
Knowledge Graphs

Some content from CS224W @ Stanford

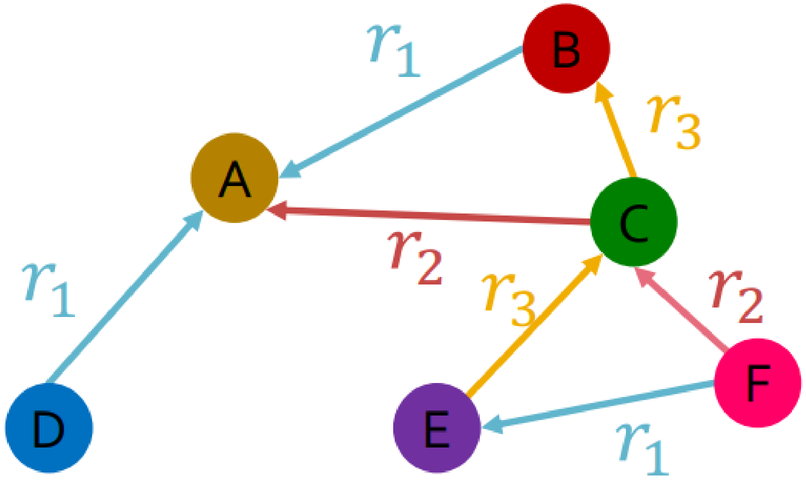
Graph:

An abstract data structure used to represent a set of objects where some pairs of the objects are connected by links.



Heterogeneous Graphs:

A Graph with multiple relation types.



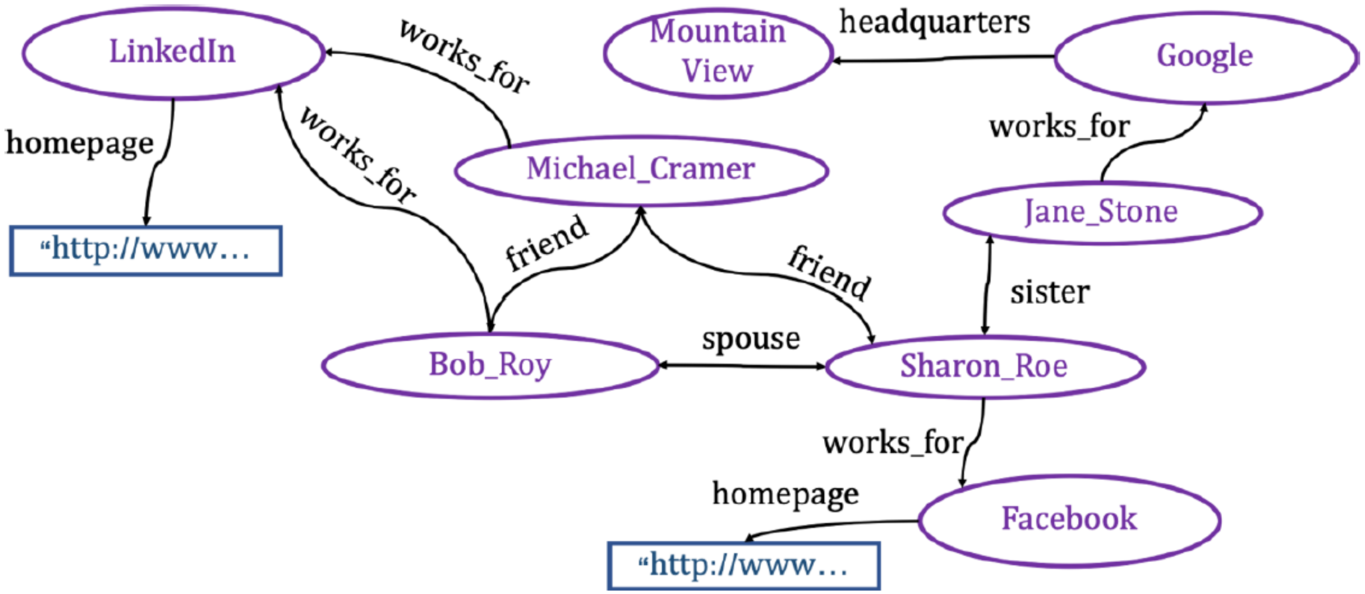
Introduction - Definition

Knowledge Graphs

A structured way to represent knowledge in a graph format. A kind of labeled, multirelational, directed graph, heterogeneous.

Nodes: Entities

Edges: capture relationships between entities



Knowledge Graphs - Examples

Widely used in search engines, recommendation systems, scientific research (biology, chemistry etc)

Examples:
- Google Knowledge Graph
- Amazon Product Graph
- Facebook Graph API
- LinkedIn Knowledge Graph
- DBpedia
- Gene Ontology
- …

Knowledge Graphs - Representation

Triples: (subject, predicate, object) / (head, relationship, tail)

Subject: The entity or the concept this triple is about

Predicate: The relationship and/or property that connects the subject to the object

Object: The entity or concept that is linked to the subject through the predicate.

Example: ("Leonardo da Vinci", "painted", "Mona Lisa")

Knowledge Graphs – Resource Description Framework (RDF)

RDF Node:

1. *Uniform resource identifier(URI)* with an optional identifier.
   Example: *https://example.com/path/resource.txt #fragment*
2. *Literal*: literal values such as strings and integers

RDF namespaces:

foaf: http://xmlns.com/foaf/0.1/

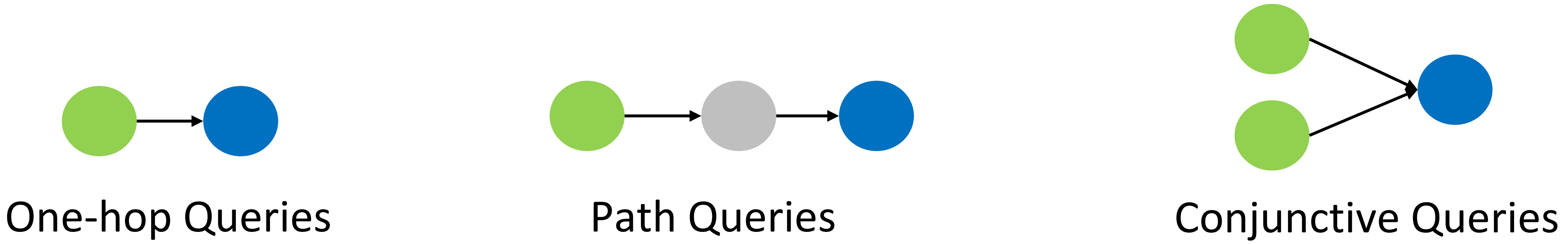
foaf:name: http://xmlns.com/foaf/ 0.1/name

rdfs: http://www.w3.org/2000/01/rdf-schema#

Accessing Knowledge Graph

Access data and present them to users in response to some kind of query

Query Types	Examples
One-hop Queries	What papers cite the paper titled 'Improvements in Machine Learning Techniques'?
Path Queries	Find the sequence of citations leading from 'Improvements in Machine Learning Techniques' to papers that cite it, and then to papers that cite those papers
Conjunctive Queries	Find all papers that cite 'Improvements in Machine Learning Techniques' and are published in the 'Journal of Advanced Computing' after 2022



Structured Querying - SPARQL

The official World Wide Web Consortium (W3C) standard for querying and extracting information from RDF graphs.

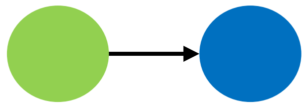
A **basic SPARQL query** is a graph pattern**.**

Graph Pattern: Triple patterns. Ordinary triples, each pattern consists of subject, predicate, and object, but each of these elements may now be a variable.

Example: *?name foaf:name 'John'*

Accessing Knowledge Graph

```
PREFIX      foaf: http://xmlns.com/foaf/0.1/
SELECT      ?person, ?homepage
FROM        http://example.org/dataset.rdf
WHERE       ?person a foaf:Person ;
             foaf:homepage ?homepage .
ORDER BY ?homepage DESC
LIMIT 5
```



One-hop Queries

PREFIX: Declare a shorthand (the "prefix") for a URL namespace.
SELECT: Similar to the SELECT often found in SQL queries.
FROM: Specify the data set over which the query will be executed.
WHERE: Where the conditions are imposed. Here, the conditions are graph patterns.
Other possible aggregation expressions: **AVG**, **MAX,MIN, COUNT, SUM**…

Anatomy of a Query

PREFIX: foaf: **PREFIX**: rdfs: …	Declare Prefix shortcuts (optional)
SELECT …	Query result clause
FROM <>	Define the dataset

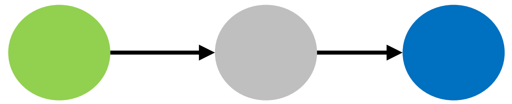
WHERE {
…
}
	Query Pattern, can embed subqueries

GROUP BY… **HAVING…** **ORDER BY…** **LIMIT…** **OFFSET…** **VALUES…**	Query Modifiers

Accessing Knowledge Graph

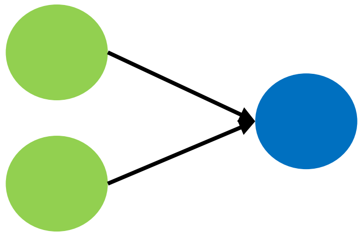
Answer Queries in SPARQL

Path Queries:
Link up one-hop SPARQL queries



Path Queries

Conjunctive Queries:
Link up one-hop SPARQL queries, utilize subqueries



Conjunctive Queries

Accessing Knowledge Graph

Public SPARQL Endpoints

Name	URL	About
DBPedia	http://dbpedia.org/sparql	Extensive RDF data from Wikipedia
DBLP	https://sparql.dblp.org/	Bibliographic data from computer science journals and conferences
Yago	https://yago-knowledge.org/sparql	Large knowledge base with general knowledge about people, cities, countries, movies, and organizations.

Accessing Knowledge Graph

SPARQL (Structured Query) Limitation

•**Dependence on Explicit Schemas**: SPARQL queries require explicit specification of patterns in the graph, challenging to construct queries for complex relationships.

•**Inability to Infer Unknown Relationship**: SPARQL operates on the static data available. It cannot infer new knowledge or predict missing relationships between entities.

•**Scalability**: efficiency performance can degrade with very large datasets or highly complex queries.

Challenges

Big Knowledge graph datasets are usually massive and incomplete (many true edges are missing).

Instance Matching: identifying different representations of the same real-world entity across or within various knowledge graphs.

Link Predicting: predicting whether a relation
(entities) based on the existing data in the gra
question answering.

Database A:
Title: "Efficient Algorithms for Graph Mining"
Authors: Jane Doe, John Smith
Published: Proceedings of the International Conference on Data Science, 2021

Database B:
Title: "Optimized Graph Mining Approaches"
Authors: J. Doe, J. A. Smith
Source: Intl. Conf. on Data Science 2021 Proceedings

Knowledge Graph Embedding

Challenges

Big Knowledge graph datasets are usually
edges are missing).

Instance Matching: identifying different represer
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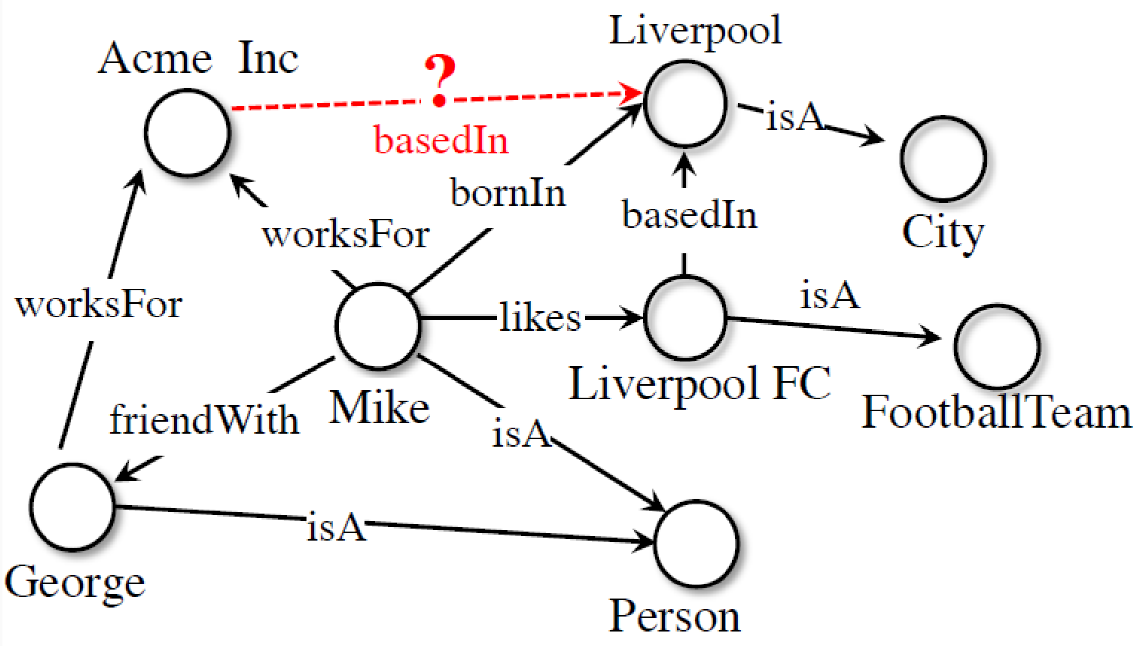


Link Predicting: predicting whether a relationship (link) should exist between two nodes (entities) based on the existing data in the graph. Crucial for content recommendation and question answering.

Knowledge Graph Embedding

Knowledge Graph Embedding (KGE)

KGE techniques transform entities and relationships of a knowledge graph into continuous vector spaces.

This representation enables the application of machine learning models to predict missing relationships, cluster entities, and perform similarity searches.

Knowledge Graph Embedding (KGE)

Given triples *(head, relation, tail)* – *($\hbar$, r, t)*

Key Idea:

- Model entities and relations in embedding space $\mathbb{R}^d$
- Given a triple *($\hbar$, r, t),* the goal is that the embedding of *($\hbar$, r)* should be close to the embedding of *t.*

Questions:

- How to embed $\hbar$, *r*?
- How to define a score f_r***(h,t)*** so that it is high if *($\hbar$, r, t)* exists, else low?

Knowledge Graph Embedding

Different KGE Models

- Based on different **geometric intuitions**

- Capture different **types of relations (**different **expressivity**)
  - **Symmetry :** `<Alice marriedTo Bob>, <Bob marriedTo Alice>`
  - **Asymmetry:** `<Alice childOf Jack>`
  - **Inversion:** `<Alice childOf Jack>, <Jack fatherOf Alice>`
  - **Composition:** `<Alice childOf Jack>, <Jack siblingOf Mary>, <Alice nieceOf Mary>`
  - **1-to-N:** `<Alice, childof Jack>, <Alice, childof, Flora>`

Common KGE Models

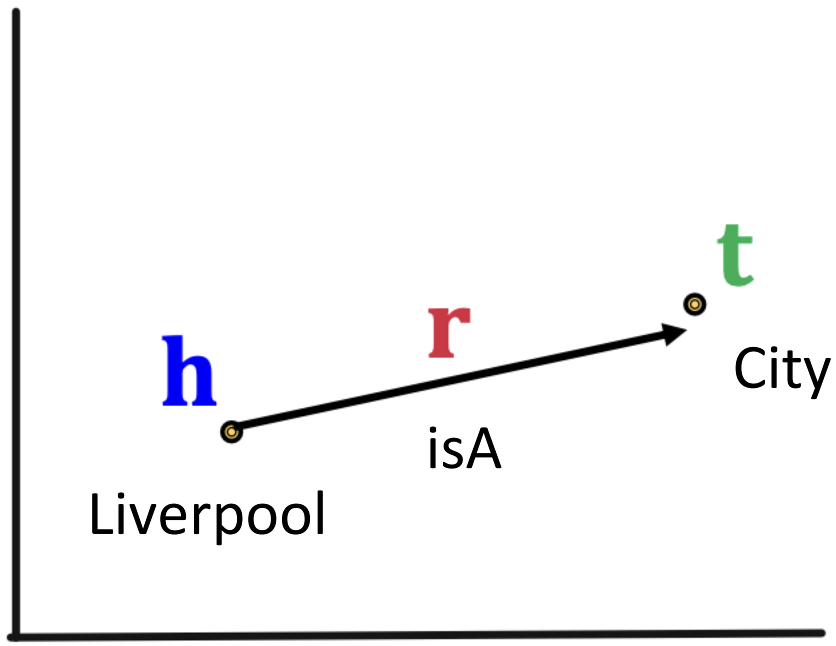
Model	Score	Embedding	Sym.	Antisym.	Inv.	Compos.	1-to-N
TransE	$-\|\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|\|$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✗	✓	✓	✓	✗
TransR	$-\|\|\boldsymbol{M}_r\mathbf{h}+\mathbf{r}-\boldsymbol{M}_r\mathbf{t}\|\|$	$\mathbf{h},\mathbf{t}\in\mathbb{R}^k$, $\mathbf{r}\in\mathbb{R}^d$, $\boldsymbol{M}_r\in\mathbb{R}^{d\times k}$	✓	✓	✓	✓	✓
DistMult	$<\mathbf{h},\mathbf{r},\mathbf{t}>$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✓	✗	✗	✗	✓
ComplEx	$\mathrm{Re}(<\mathbf{h},\mathbf{r},\bar{\mathbf{t}}>)$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{C}^k$	✓	✓	✓	✗	✓

TransE

For a triple (h, r, t), let $\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{R}^d$ be embedding vectors.

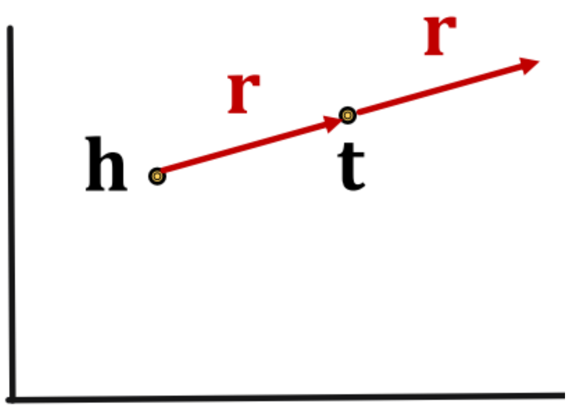
Translation: - **h + r ≈ t** if the given link exists, else **h + r ≠ t**

$$f_r(h, t) = -||\mathbf{h} + \mathbf{r} - \mathbf{t}||$$



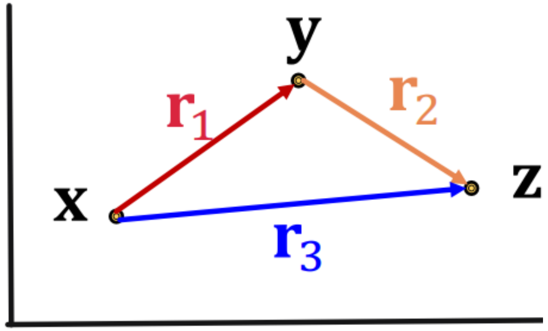
Knowledge Graph Embedding

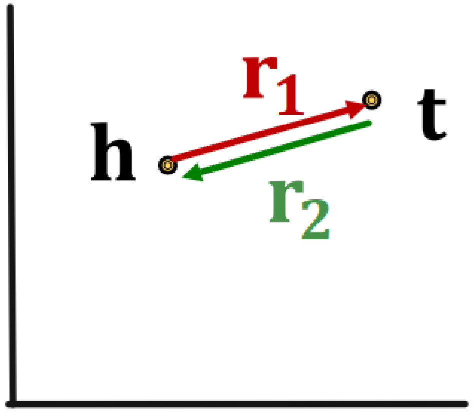
TransE Abilities



Translation: - **h + r ≈ t** if the given link exists, else **h + r ≠ t**

- **Antisymmetric Relations: h + r = t,** but **t+ r ≠ h**
- **Inverse Relations**: **h** + **r_1** = **t**, **t** + **r_2** = **h**, we can set **r_1** = -**r_2**
- **Composition Relations: $r_3 = r_1 + r_2$**





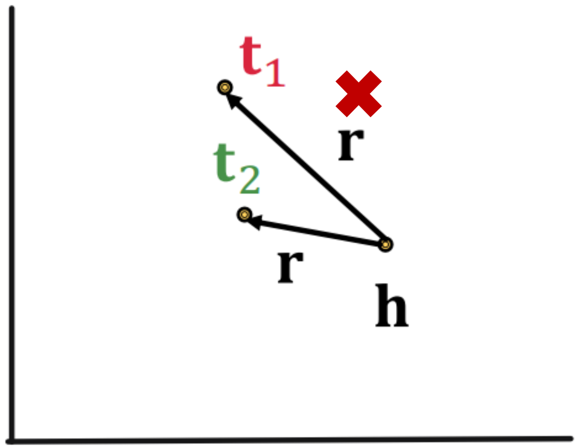
TransE Limitations

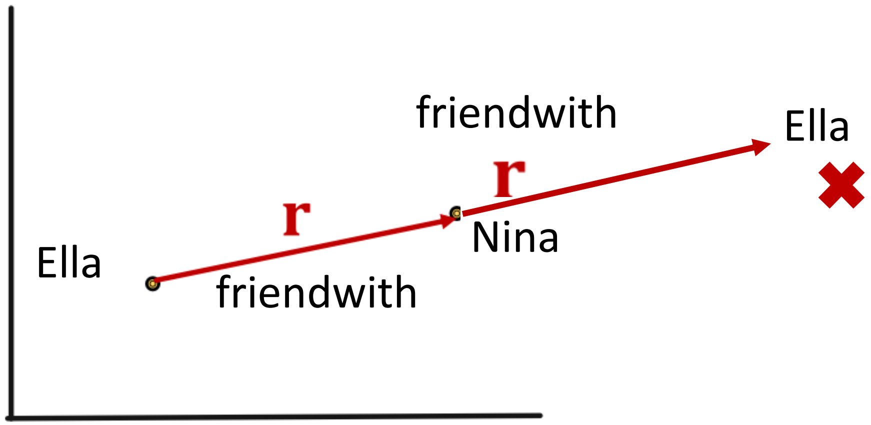
Translation: - **h + r ≈ t** if the given link exists, else **h + r ≠ t**

- ❌ **Symmetric Relations** (e.g. friend with, roommate)
- ❌ 1-N **Relations** (e.g. "studentsof")

$$\mathbf{t}_1 = \mathbf{h} + \mathbf{r} = \mathbf{t}_2$$
$$\mathbf{t}_1 \neq \mathbf{t}_2$$

contradictory!





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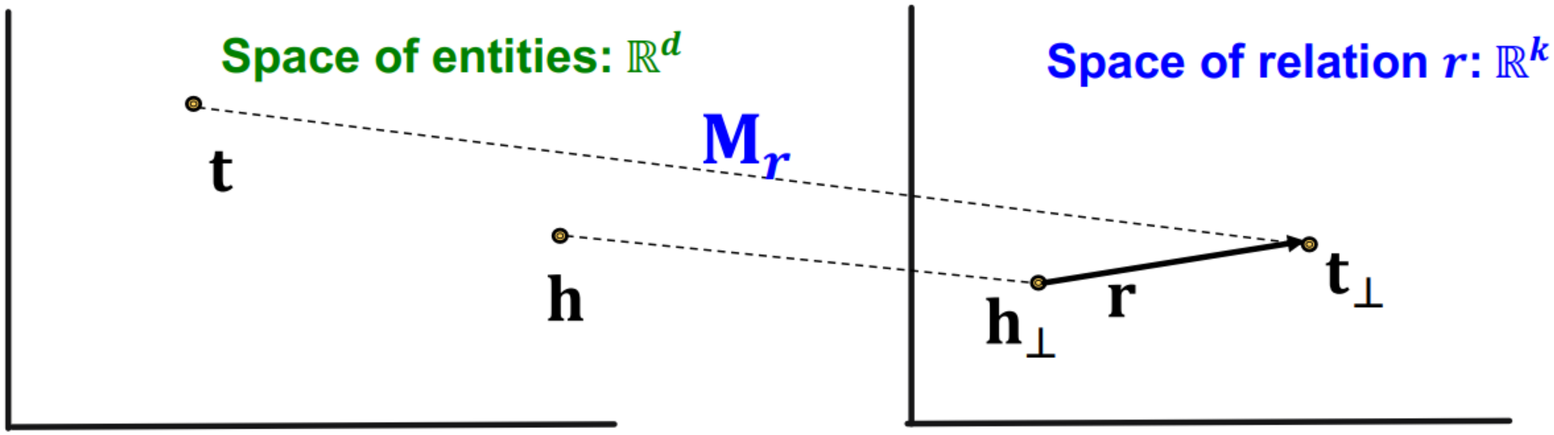
TransR	$-\|\boldsymbol{M}_r\mathbf{h}+\mathbf{r}-\boldsymbol{M}_r\mathbf{t}\|$	$\mathbf{h},\mathbf{t}\in\mathbb{R}^k$, $\mathbf{r}\in\mathbb{R}^d$, $\boldsymbol{M}_r\in\mathbb{R}^{d\times k}$	✓	✓	✓	✓	✓
DistMult	$<\mathbf{h},\mathbf{r},\mathbf{t}>$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✓	✗	✗	✗	✓
ComplEx	$\mathrm{Re}(<\mathbf{h},\mathbf{r},\bar{\mathbf{t}}>)$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{C}^k$	✓	✓	✓	✗	✓

TransR

TransR models entities as vectors in the entity space $\mathbb{R}^d$ and models each relation as vector in relation space $\mathbf{r} \in \mathbb{R}^k$ with (relation-specific) $\mathbf{M_r} \in \mathbb{R}^{kxd}$ as the projection matrix.

$$\mathbf{h}_\perp = \mathbf{M}_r\mathbf{h}, \ \mathbf{t}_\perp = \mathbf{M}_r\mathbf{t}$$

$$f_r(h,t) = -||\mathbf{h}_\perp + \mathbf{r} - \mathbf{t}_\perp||$$

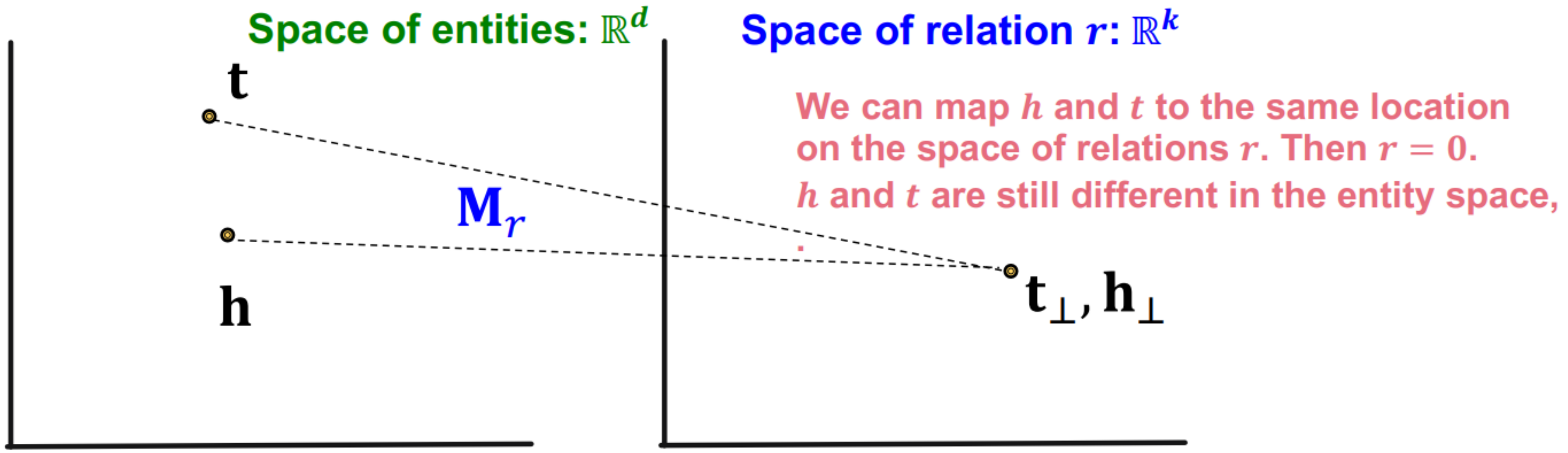


TransR Abilities

TransR models entities as vectors in the entity space $\mathbb{R}^d$ and models each relation as vector in relation space $\mathbf{r} \in \mathbb{R}^k$ with $\mathbf{M}_r \in \mathbb{R}^{k \times d}$ as the projection matrix.

Symmetric Relations :
(roommate, friendwith**)**

$$\mathbf{r} = 0,\ \mathbf{h}_\perp = \mathbf{M}_r\mathbf{h} = \mathbf{M}_r\mathbf{t} = \mathbf{t}_\perp$$



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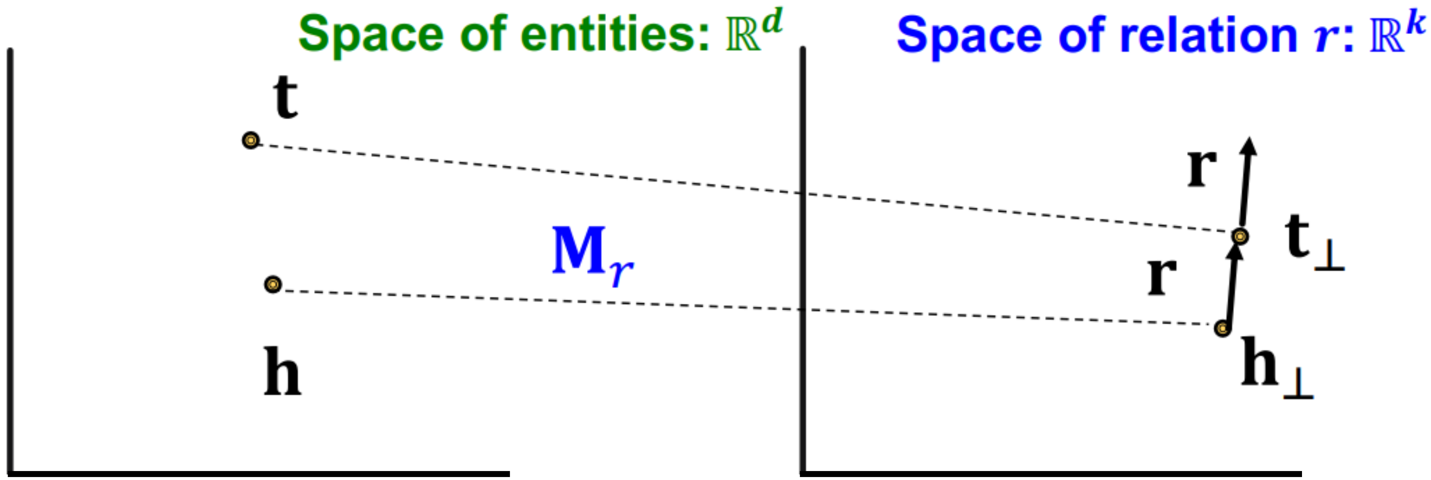
Antisymmetric Relations:

$\mathbf{r} \neq 0, \mathbf{M}_r\mathbf{h} + \mathbf{r} = \mathbf{M}_r\mathbf{t},$

Then $\mathbf{M}_r\mathbf{t} + \mathbf{r} \neq \mathbf{M}_r\mathbf{h}$ ✓



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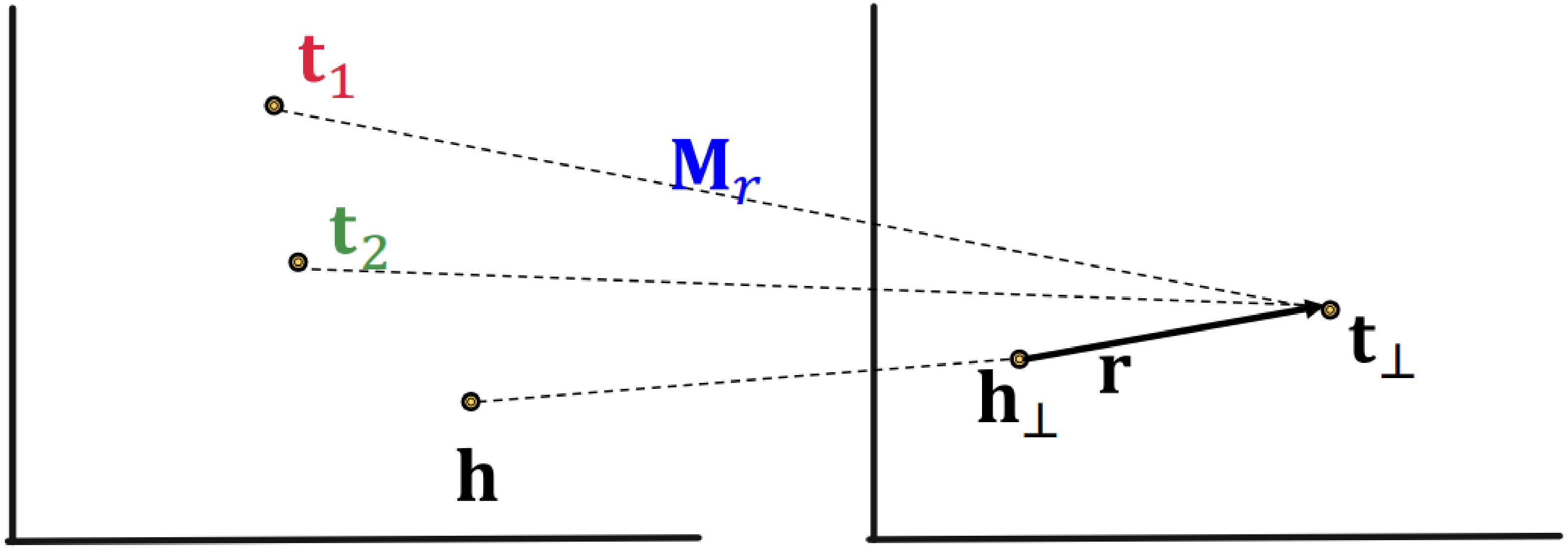
- **1-to-N Relations**: Learn M_r so that (Studentsof)

$$\mathbf{t}_\perp = \mathbf{M}_r\mathbf{t}_1 = \mathbf{M}_r\mathbf{t}_2$$



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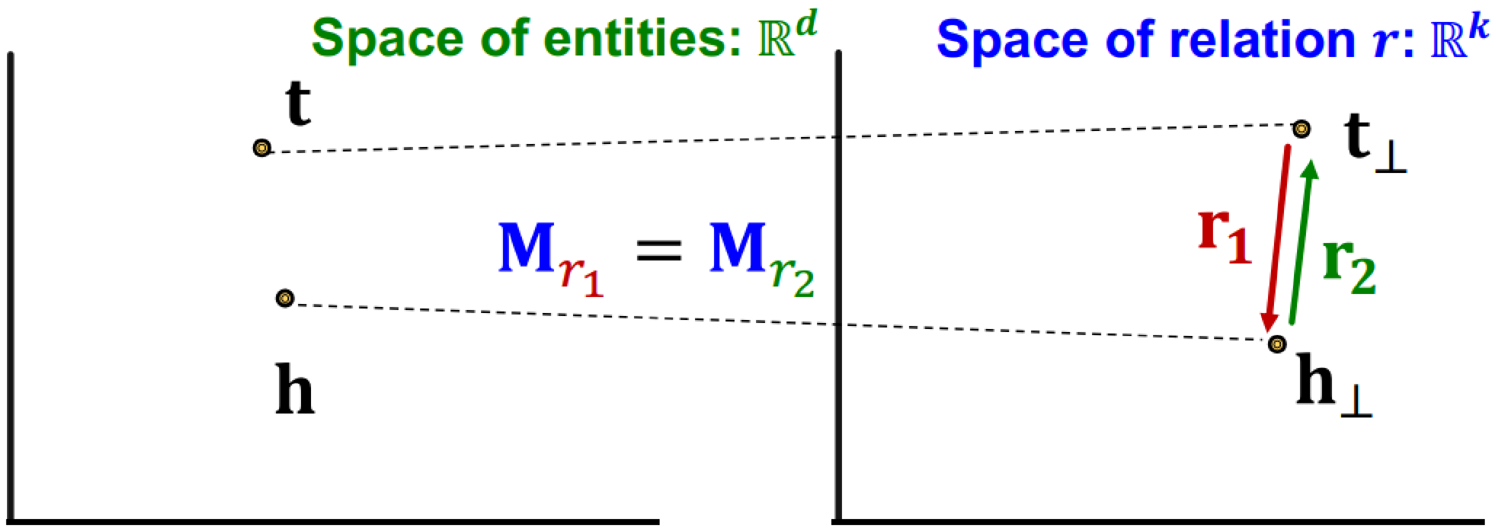
TransR models entities as vectors in the entity space $\mathbb{R}^d$ and models each relation as vector in relation space $\mathbf{r} \in \mathbb{R}^k$ with $\mathbf{M}_r \in \mathbb{R}^{k \times d}$ as the projection matrix.

✓ **Inverse Relations**: (employer, employee)

$$\mathbf{r}_2 = -\mathbf{r}_1, \mathbf{M}_{r_1} = \mathbf{M}_{r_2}$$

$$\mathbf{M}_{r_1}\mathbf{t} + \mathbf{r}_1 = \mathbf{M}_{r_1}\mathbf{h}$$

$$\mathbf{M}_{r_2}\mathbf{h} + \mathbf{r}_2 = \mathbf{M}_{r_2}\mathbf{t} \checkmark$$



TransR Abilities

TransR models entities as vectors in the entity space $\mathbb{R}^d$ and models each relation as a vector in relation space $\mathbf{r} \in \mathbb{R}^k$ with $\mathbf{M}_r \in \mathbb{R}^{k \times d}$ as the projection matrix.

Composition Relations

High-level intuition: TransR models a triple with linear functions. Linear functions are chainable!

Common KGE Models

Model	Score	Embedding	Sym.	Antisym.	Inv.	Compos.	1-to-N
TransE	$-\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✗	✓	✓	✓	✗
TransR	$-\|\boldsymbol{M}_r\mathbf{h}+\mathbf{r}-\boldsymbol{M}_r\mathbf{t}\|$	$\mathbf{h},\mathbf{t}\in\mathbb{R}^k$, $\mathbf{r}\in\mathbb{R}^d$, $\boldsymbol{M}_r\in\mathbb{R}^{d\times k}$	✓	✓	✓	✓	✓
DistMult	$<\mathbf{h},\mathbf{r},\mathbf{t}>$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✓	✗	✗	✗	✓
ComplEx	$\mathrm{Re}(<\mathbf{h},\mathbf{r},\bar{\mathbf{t}}>)$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{C}^k$	✓	✓	✓	✗	✓

Knowledge Graph Embedding

Knowledge Graph Embedding (KGE)

•**Inference and Prediction**: KGE can infer potential relationships not explicitly present in the data, potential for discovering hidden patterns or predicting missing links.

•**Scalability and Efficiency**: KGE models, once trained, can efficiently handle queries on large graphs since they work in a compact, continuous vector space.

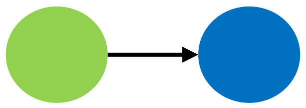
•**Generalization**: Embeddings can capture and generalize patterns in the data, enabling queries that are more about the nature or category of relationships rather than specific entities.

Knowledge Graph Embedding

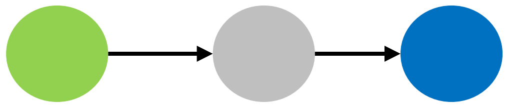
Answer Queries on KGE

Especially useful when KG is **massive** and **incomplete**: many relations between entities are missing.

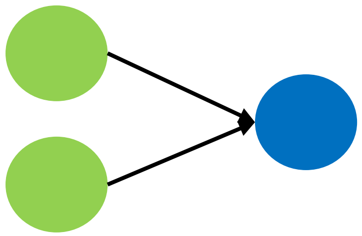
If we first do KG completion – the completed KG is a dense graph.
Time complexity of traversing a dense KG is exponential of the path length.



One-hop Queries

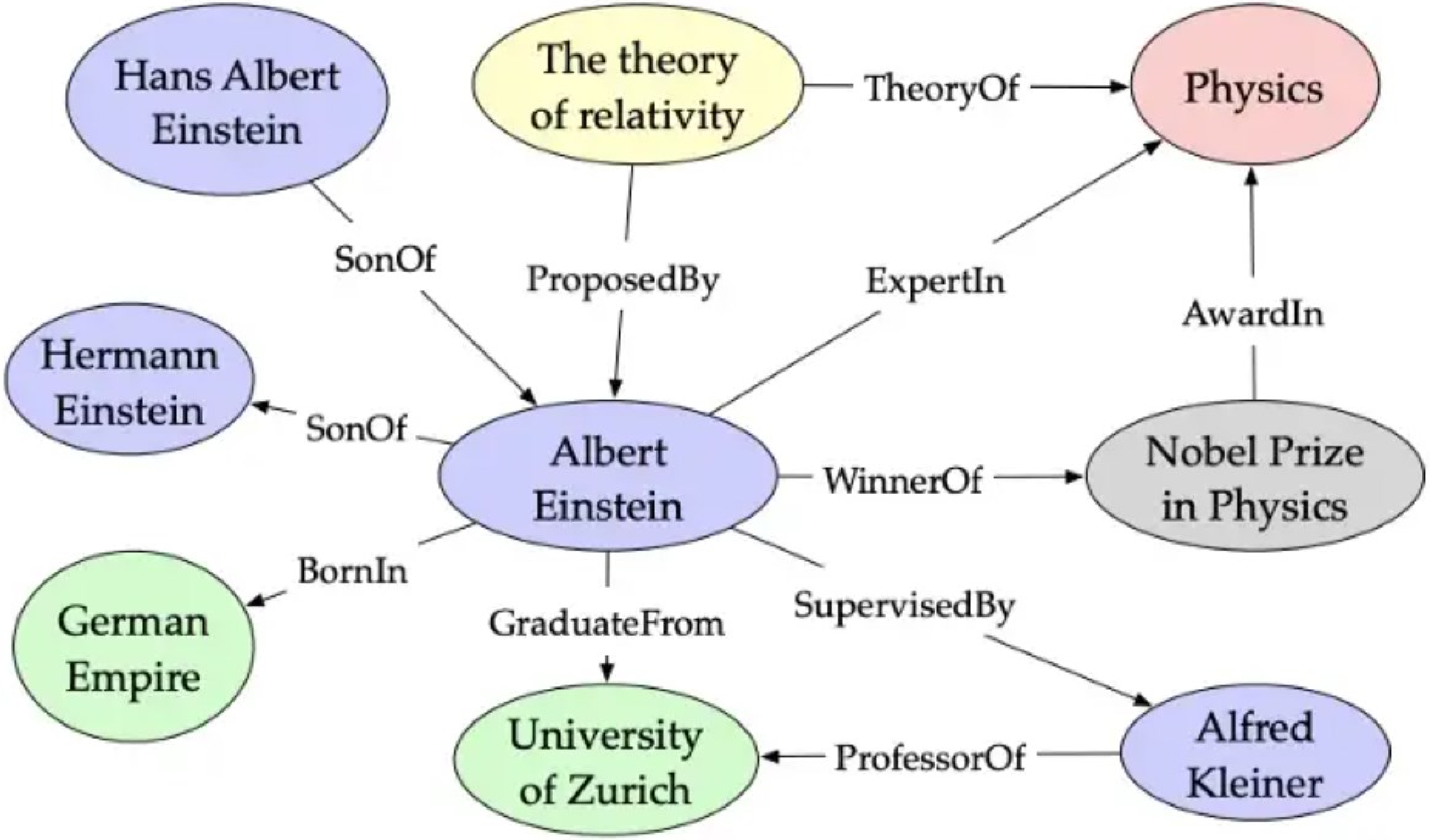


Path Queries



Conjunctive Queries

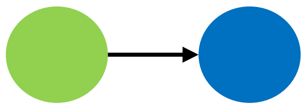
Example Graph



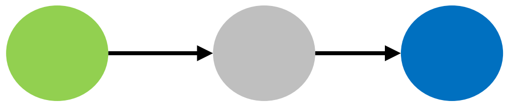
Answer Queries on KGE

Idea: Map queries into embedding space, traverse KG in the embedding space.

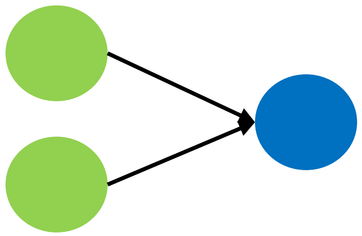
- **One-hop Query**: can be seen as a kind of link prediction - is (h,r,t) in the KG? Namely, is t an answer to (h,r)?
- **Path Query**: chained one-hop queries.
- **Conjunctive Queries:** complex queries, Query2Box



One-hop Queries



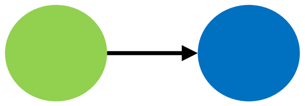
Path Queries



Conjunctive Queries

Knowledge Graph Embedding

Answer One-hop Queries on KGE

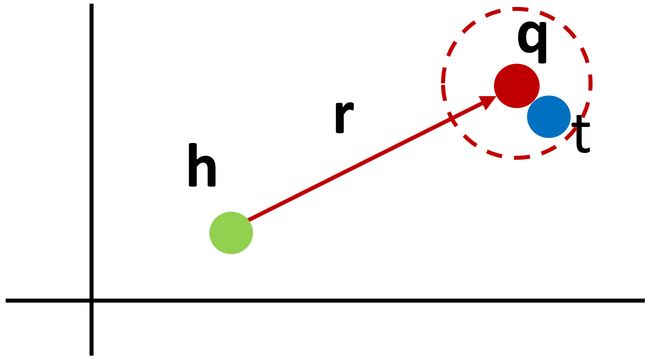


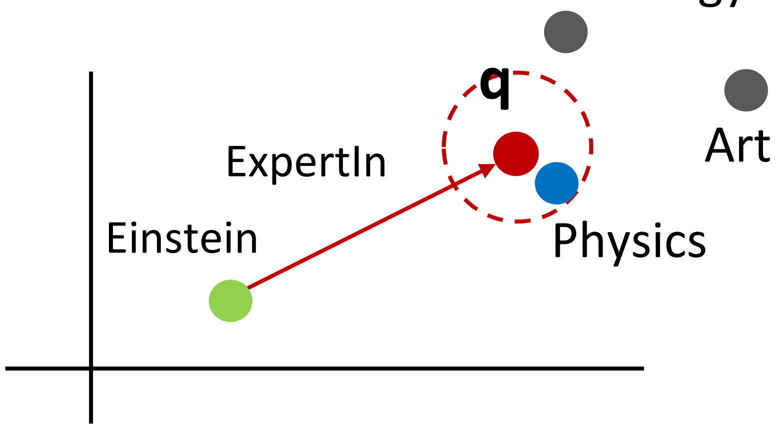
Idea: Embed query into a single point in the Euclidean space, the answer nodes are close to the query in this space.

TransE:

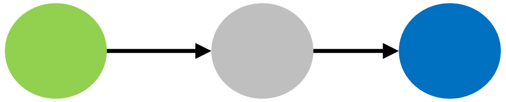
Query embedding q = h + r should be close to the **answer embedding t**

$$f_q(t) = -||\mathbf{q} - \mathbf{t}||$$





Answer Path Queries on KGE

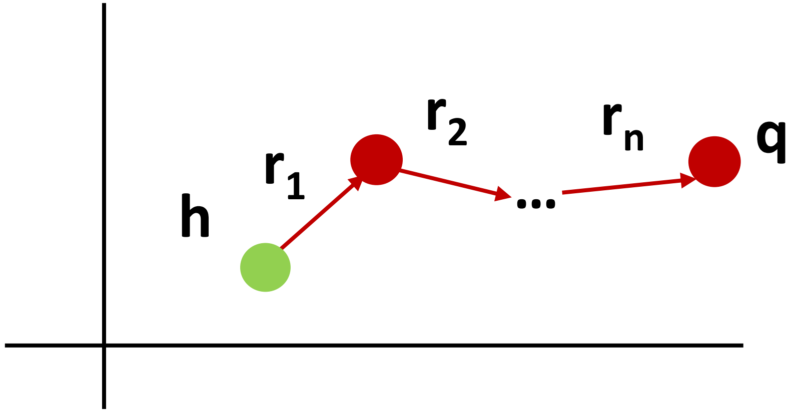


Path Queries

Idea: Chained one-hop queries, the embedding process use addition of relation embeddings (vectors).

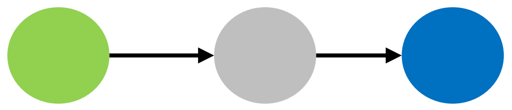
TransE:

Query embedding $q = h + r_1 + r_2 + \ldots + r_n$



Knowledge Graph Embedding

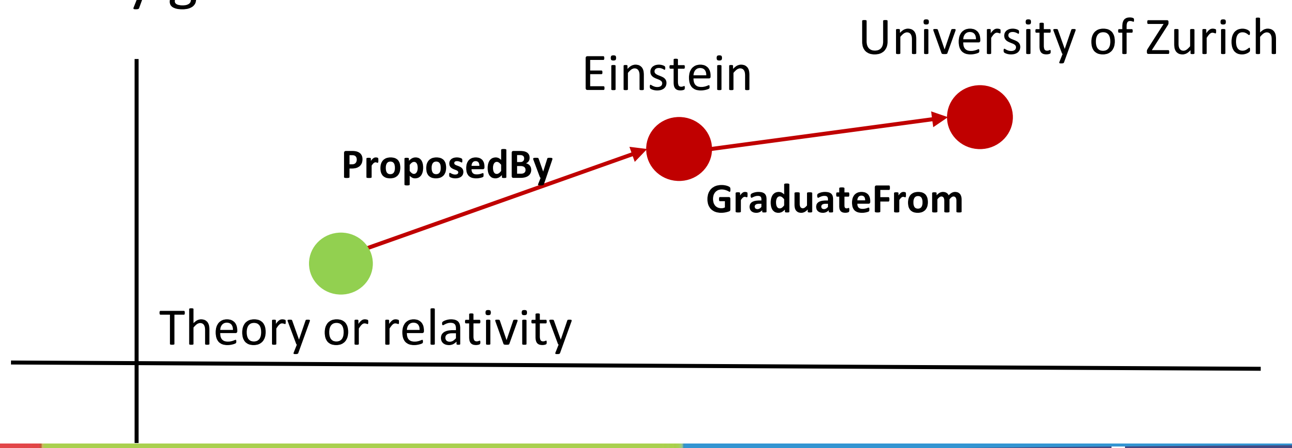
Answer Path Queries on KGE



Idea: Chained one-hop queries, the embedding process use addition of relation embeddings (vectors).

Query embedding $\mathbf{q} = \mathbf{h} + \mathbf{r}_1 + \mathbf{r}_2 + \ldots + \mathbf{r}_n$

Example Question:
Where did the inventor of the theory of relativity graduate?



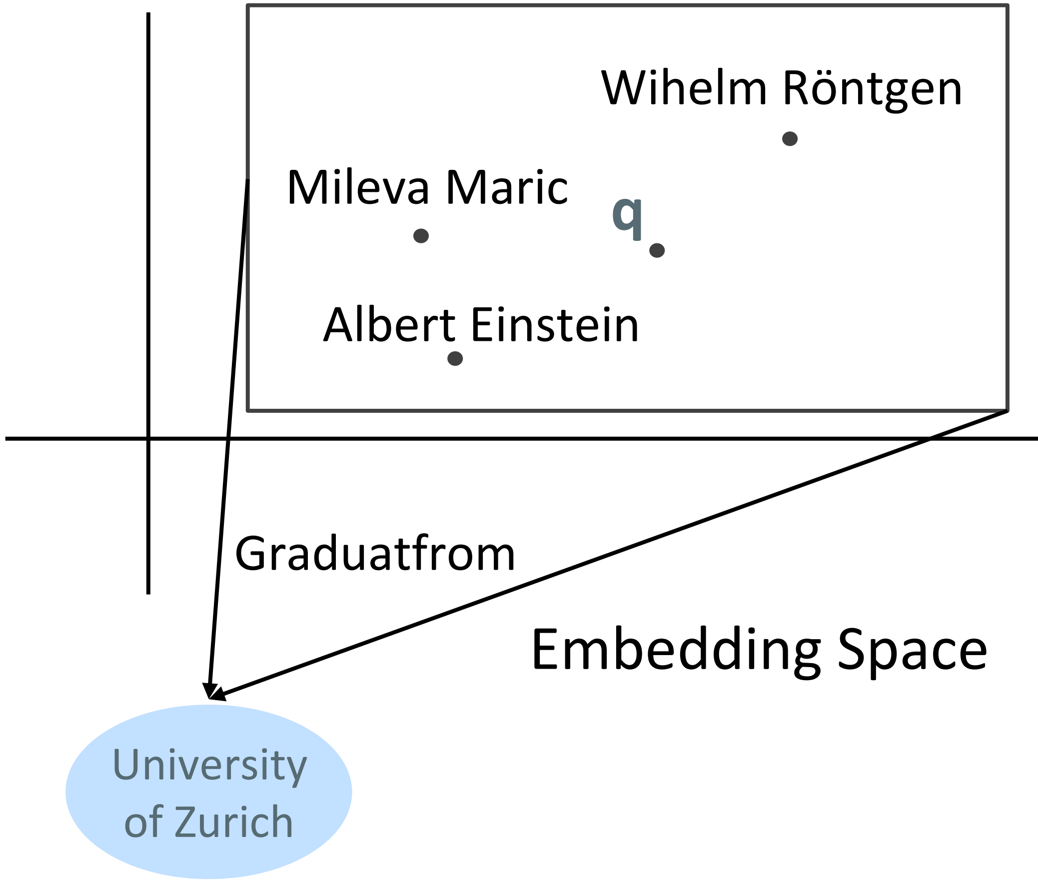
Knowledge Graph Embedding

Answer Conjunctive Queries on KGE

$$\mathbf{q} = (Center(q), Offset(q))$$

Idea: Embed queries with hyper-rectangles (boxes)

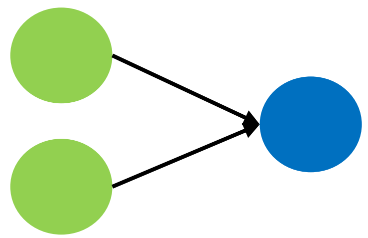
- **Entities** are seen as zero-volume boxes
- Each **relation** takes a box and produces a new box
- **Projection operator** $\mathcal{P}$ takes the current box as input and use the relation embedding to project and expand the box



Knowledge Graph Embedding

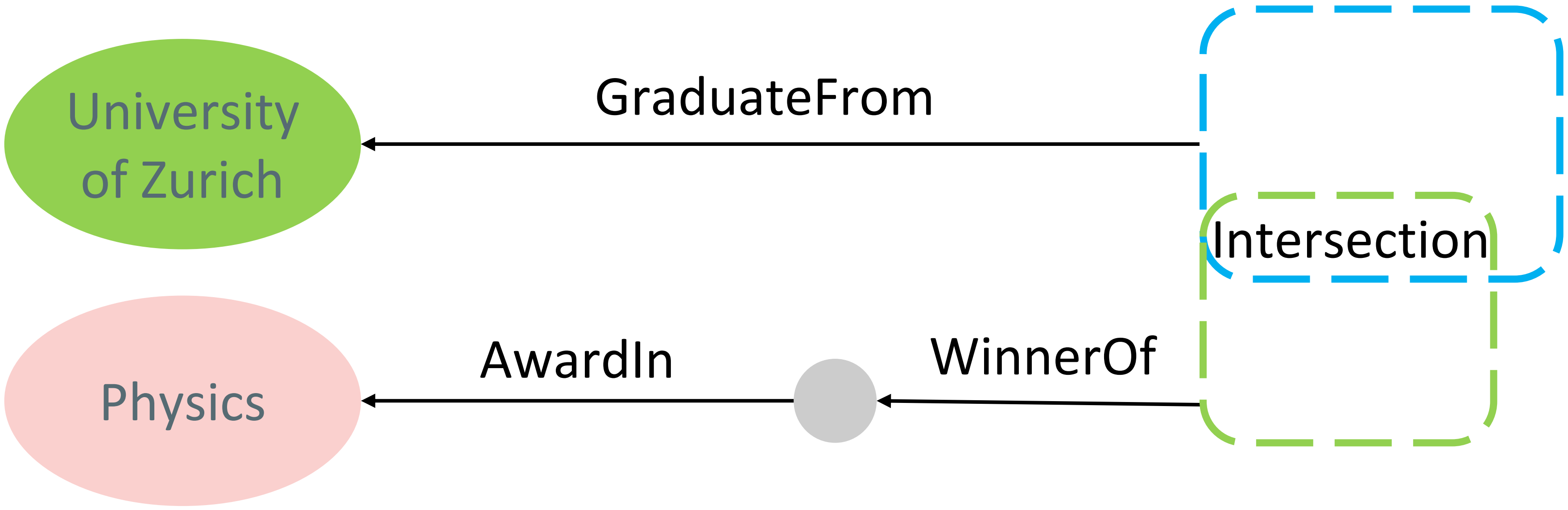
Answer Conjunctive Queries on KGE

Idea: Embed queries with hyper-rectangles (boxes). Answer of the query can be seen as the intersection of the boxes.

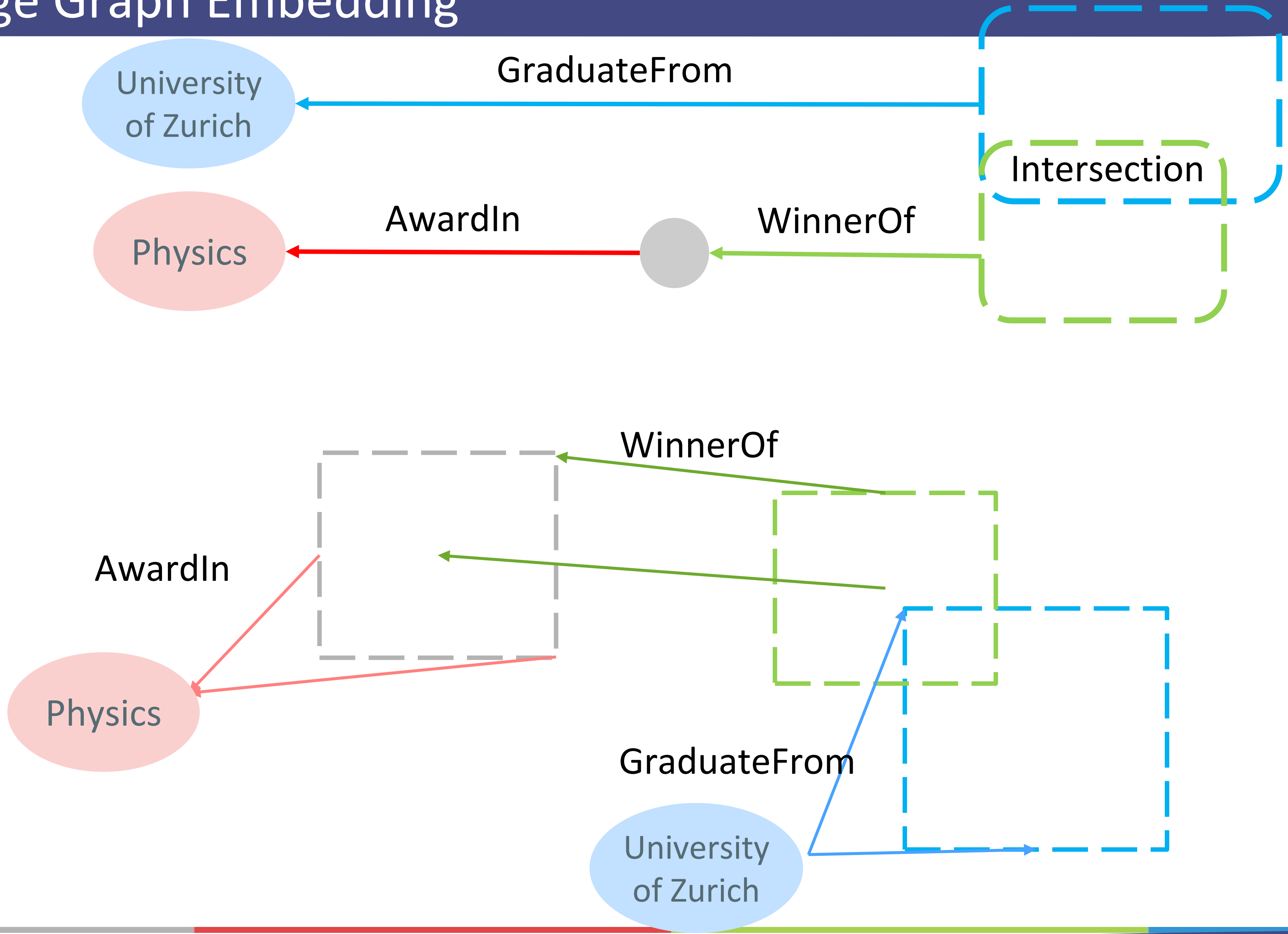


Conjunctive Queries

Example: Which Alumni of University of Zurich has won a price in Physics?



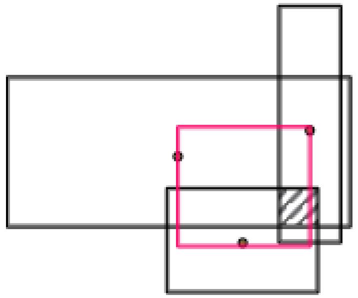
Knowledge Graph Embedding



Answer Conjunctive Queries on KGE

Geometric Intersection Operator $\mathcal{I}$ take multiple boxes as input and produce the intersection box.

Center: weighted sum of the input box centers



(element-wise product)

$$Cen(q_{inter}) = \sum_i \mathbf{w}_i \odot Cen(q_i)$$

$$\mathbf{w}_i = \frac{\exp(f_{cen}(Cen(q_i)))}{\sum_j \exp(f_{cen}(Cen(q_j)))}$$

$Cen(q_i) \in \mathbb{R}^d$

$\mathbf{w}_i \in \mathbb{R}^d$

Answer Conjunctive Queries on KGE

Geometric Intersection Operator $\mathcal{I}$ take multiple boxes as input and produce the intersection box.

Offset:

- Minimum of the offset of the input box
- A new function f_{off}: a neural network trained to extract the representation of the input boxes with a sigmoid function to guarantee shrinking.

$$
\begin{aligned}
&Off(q_{inter}) \\
&= \min\big(Off(q_1), \dots, Off(q_n)\big) \\
&\odot \sigma(f_{off}(Off(q_1), \dots, Off(q_n)))
\end{aligned}
$$

Summary

Autonomous Agent

- Definition
- Characteristic

Knowledge Graph

- Definition and Terms
- Modelling and Representing
- SPARQL
- Knowledge Graph Embedding

Dialogue Tree

- Definition
- Comparison with LLMs
- Rasa

Reading List

Autonomous Agents

- The Handbook on Socially Interactive Agents: 20 years of Research on Embodied Conversational Agents, Intelligent Virtual Agents, and Social Robotics Volume 1 and 2

Dialogue Tree

- Twine: https://twinery.org/reference/en/index.html
- Rasa: https://learning.rasa.com/

Knowledge Graph:

- Knowledge Graphs: Fundamentals, Techniques, and Applications by Mayank Kejriwal
- CS224W Stanford: https://web.stanford.edu/class/cs224w/
- https://vldb.org/workshops/2024/proceedings/LLM+KG/LLM+KG-9.pdf
- https://dl.acm.org/doi/10.1109/TKDE.2024.3352100